The leaflet document class

Jürgen Schlegelmilch Hubert Gäßlein Rolf Niepraschk
Walter Schmidt

2020/11/06

Contents

1 Usage **1**

2 Implementation **1**
2.1 Test whether the LaTeX format is new enough 1
2.2 Saving some internal macros. 1
2.3 Option processing 1
2.4 Changing the class properties 7
2.5 Changing the page dimensions 8
2.6 Additional user macros 9
2.7 Page construction and output 10
2.7.1 Intercepting the small pages 10
2.7.2 Reassembling the small pages 11

1 Usage

Process the file `leaflet-manual.tex` with LaTeX to see details of usage and layout.

2 Implementation

```
1 ⟨*class⟩
```

2.1 Test whether the LaTeX format is new enough

We use the 'shipout/before' hook.

```
2 \providecommand\IfFormatAtLeastTF{\@ifl@t@r\fmtversion}
3 \IfFormatAtLeastTF{2020/10/01}{}{%
4   \ClassError{leaflet}{%
5     LaTeX version equal to or newer \MessageBreak
6     than '2020/10/01' is required%
7   }{%
8     Renew your TeX installation.%
9   }%
10 }
```

2.2 Saving some internal macros. . .

```
11 \let\LL@begindvi\@begindvi
```

2.3 Option processing

```
12 \@ifundefined{iflandscape}{\newif\iflandscape}{}%
13 \@ifundefined{iftumble}{\newif\iftumble}{}%
14 \@ifundefined{iftwopart}{\newif\iftwopart}{}%
15 \@ifundefined{iffoldcorr}{\newif\iffoldcorr}{}%
16 \newlength\LL@tempdima
17 \newcommand\LL@debug@info[1]{}%
```

```
\DeclareOption{dvips}{\PassOptionsToPackage{\CurrentOption}{graphics}}
\DeclareOption{pdftex}{\PassOptionsToPackage{\CurrentOption}{graphics}}
\DeclareOption{vtex}{\PassOptionsToPackage{\CurrentOption}{graphics}}
\DeclareOption{dvipdfm}{\PassOptionsToPackage{\CurrentOption}{graphics}}
\DeclareOption{twoside}{\OptionNotUsed}
\DeclareOption{twocolumn}{\OptionNotUsed}
\DeclareOption{landscape}{\landscapetrue}
\DeclareOption{portrait}{\landscapefalse}
\DeclareOption{debug}{\let\LL@debug@info\typeout}
\DeclareOption{twopart}{\twoparttrue}
\DeclareOption{notwopart}{\twopartfalse}
\DeclareOption{nospecialtricks}{%
  \AtEndOfClass{%
    \ifLL@combine
      \let\immediate\@@@immediate\let\write\@@@write
      \let\openout\@@@openout\let\closeout\@@@closeout
      \let\special\@@@special\let\@@@exec@outs\relax
    \fi}}
```

`\LL@setPaperSize` Some paper sizes are not supported by the article class. Hence this work-around. These page dimensions *must* be given in landscape orientation!

```
\newcommand*\LL@setPaperSize{}
\DeclareOption{a3paper}{\def\LL@setPaperSize{%
  \paperwidth=420mm\paperheight=297mm\relax}}%
\@ifdefinable\ifLL@combine{\newif\ifLL@combine}
\DeclareOption{combine}{\LL@combinetrue}
\DeclareOption{nocombine}{\LL@combinefalse}
```

`\LL@selectOutput`

```
\newcommand*\LL@selectOutput{}
\DeclareOption{frontside}{\def\LL@selectOutput#1#2{#1}}
\DeclareOption{backside}{\def\LL@selectOutput#1#2{#2}}
\DeclareOption{bothsides}{\def\LL@selectOutput#1#2{#1#2}}

\DeclareOption{tumble}{\tumbletrue}
\DeclareOption{notumble}{\tumblefalse}
```

`\LL@foldmark` Drawing code for the fold mark. (Will be called within a `picture` environment.)

```
\newcommand*\LL@foldmark{}
\DeclareOption{foldmark}{%
  \def\LL@foldmark{%
    \begingroup
      \linethickness{\foldmarkrule}%
      \@tempdima=\dimexpr\paperheight-\LL@tmargin
      \put(0,\@tempdima){%
        \line(0,-1){\foldmarklength}}%
    \endgroup}%
}

\DeclareOption{nofoldmark}{\def\LL@foldmark{}}%


\DeclareOption{foldcorr}{\foldcorrtrue}
\DeclareOption{nofoldcorr}{\foldcorrfalse}

\newcommand*\LL@toomanypages[2]{}
\DeclareOption{draft}{\PassOptionsToClass{\CurrentOption}{article}%
  \AtEndOfClass{%
    \def\LL@toomanypages#1#2{%
        \ClassWarningNoLine{leaflet}{#1.\MessageBreak#2}}%
  }%
}
\DeclareOption{final}{\PassOptionsToClass{\CurrentOption}{article}%
  \AtEndOfClass{%
    \ifLL@combine
```

```
      \def\LL@toomanypages#1#2{%
        \ClassError{leaflet}{#1}{#2.}}%
    \else
      \def\LL@toomanypages#1#2{%
        \ClassWarningNoLine{leaflet}{#1.\MessageBreak#2}}%
    \fi
  }%
}
\DeclareOption*{\PassOptionsToClass{\CurrentOption}{article}}
\PassOptionsToClass{landscape,a4paper}{article}
\ExecuteOptions{tumble,foldmark,bothsides,combine,landscape,notwopart}
\ProcessOptions\relax
\ifLL@combine
```

`\LL@rotate@I` `\LL@rotate@II` Rotates the output pages.

```
  \newcommand*\LL@rotate@I{}\newcommand*\LL@rotate@II{}%
  \iflandscape
    \def\LL@rotate@I#1{#1}%
    \iftumble
      \def\LL@rotate@II#1{\rotatebox[origin=c]{180}{#1}}%
    \else
      \def\LL@rotate@II#1{#1}%
    \fi
  \else
    \def\LL@rotate@I#1{\rotatebox[origin=c]{90}{#1}}%
    \iftumble
      \def\LL@rotate@II#1{\rotatebox[origin=c]{270}{#1}}%
    \else
      \def\LL@rotate@II#1{\rotatebox[origin=c]{90}{#1}}%
    \fi
  \fi
```

`\immediate` `\write` `\openout` `\closeout` In the following we redefine the TEX primitives `\immediate`, `\write`, `\openout`, and `\closeout`. All of them not "immediate" executed statements are added to the macro `\@@@pending@outs` (comma list). So especially the `\write` statements independent of TEX's `\shipout` run at a convenient time. For VTEX we must keep in mind that `\immediate\special` exist.

```
  \def\@@@pending@outs{}\let\@@@immediate\immediate
  \let\@@@write\write \let\@@@special\special
  \let\@@@openout\openout \let\@@@closeout\closeout
  \def\immediate{%
    \let\write\immediate@write%
    \let\openout\immediate@openout%
    \let\closeout\immediate@closeout%
    \let\special\immediate@special}%
  \def\reset@immediate{%
    \let\write\pending@write%
    \let\openout\pending@openout%
    \let\closeout\pending@closeout%
    \let\special\@@@special}%
  \long\def\pending@write#1#{\pending@@write{#1}}
  \def\immediate@write{%
    \reset@immediate\@@@immediate\@@@write}%
  \def\immediate@openout{%
    \reset@immediate\@@@immediate\@@@openout}%
  \def\immediate@closeout{%
    \reset@immediate\@@@immediate\@@@closeout}%
  \def\immediate@special{%
    \reset@immediate\@@@immediate\@@@special}%
  \let\write\pending@write
  \let\openout\pending@openout
  \let\closeout\pending@closeout
```

\@dummy@whatsit Makes our redefined primitives more like "whatsits" (???). '\special{}' is also possible for pdfTEX but we get a lot of strange warnings ("Non-PDF special ignored!") in this case.

```
\def\@dummy@whatsit{\special{}}
\begingroup\@ifundefined{pdfoutput}%
{\endgroup}
{\endgroup
 \ifnum\pdfoutput>\z@\def\@dummy@whatsit{\pdfliteral{}}\fi}

\begingroup\expandafter\expandafter\expandafter\endgroup
\expandafter\ifx\csname eTeXversion\endcsname\relax
  %%% Test is from Markus Kohm (d.c.t.t, 29 Jun 2004)
  \ClassWarningNoLine{leaflet}{%
  ************************************\MessageBreak
  * It's very recommended to use eTeX  \MessageBreak
  * with this package!                 \MessageBreak
  ************************************}%
```

\pending@@write Doesn't work correctly without any \mark or \special or \pdfliteral command inside.

```
  \long\def\pending@@write#1#2{%
    \@dummy@whatsit
    \g@addto@macro\@@@pending@outs{\@@@immediate\@@@write\number#1{#2},}}%
```

\pending@openout

```
  \def\pending@openout#1 {%
    \@dummy@whatsit
    \g@addto@macro\@@@pending@outs{\@@@immediate\@@@openout\number#1,}}%
```

\pending@closeout

```
  \def\pending@closeout#1{%
    \@dummy@whatsit
    \g@addto@macro\@@@pending@outs{\@@@immediate\@@@closeout\number#1,}}%
```

\@@@exec@outs Note to this implementation: The out commands and the page breaks are not allways in sync, which means possible wrong page references.

```
  \newcommand*\@@@exec@outs{%
    \@@@pending@outs\gdef\@@@pending@outs{}%
    \LL@debug@info{%
      >>> execute the output commands of the current page <<<}}%

\else
  \RequirePackage{etex}
```

Allocation of a new mark class.

```
  \globmarks\@@@out@mark
  \newcounter{@@total@outs}\setcounter{@@total@outs}{0}
  \newcounter{@@last@exec}\setcounter{@@last@exec}{0}
```

\pending@@write Doesn't work correctly without any \mark or \special or \pdfliteral command inside.

```
\long\def\pending@@write#1#2{%
  \global\advance\c@@@total@outs\@ne%
  \marks\@@@out@mark{\the\c@@@total@outs}%
  \g@addto@macro\@@@pending@outs{\@@@immediate\@@@write\number#1{#2},}}%
```

\pending@openout
\pending@@openout
\pending@@@openout
\@ifequalsign

```
\def\pending@openout#1 {%
  \global\advance\c@@@total@outs\@ne%
  \marks\@@@out@mark{\the\c@@@total@outs}%
  \g@addto@macro\@@@pending@outs{\@@@immediate\@@@openout\number#1,}}%
```

\pending@closeout

```
\def\pending@closeout#1{%
  \global\advance\c@@@total@outs\@ne%
  \marks\@@@out@mark{\the\c@@@total@outs}%
  \g@addto@macro\@@@pending@outs{\@@@immediate\@@@closeout\number#1,}}%
```

`\@@@exec@outs` With the call of `\@@@exec@outs` we simulate TEX's execution of non-immediate output commands. The number of pending commands which can be executed must be decremented by `\value{@@total@outs}-\botmarks\@@@out@mark`. The remaining commands must still wait until the next output page is ready. It works only for output commands in the main text which moved to the next page but not for floating output commands.

```
\newcommand*\@@@exec@outs{%
  \begingroup
```

Pending io's = `\@tempcntb` = TOTAL-OUTS − LAST-EXEC

```
    \@tempcntb\c@@@total@outs\advance\@tempcntb-\c@@@last@exec%
    \edef\reserved@a{\botmarks\@@@out@mark}%
    \ifx\reserved@a\@empty\@tempcnta\z@\else\@tempcnta\reserved@a\fi%
    \LL@debug@info{PENDING-OUTS:\the\@tempcntb\space\space
                   TOTAL-OUTS:\the\c@@@total@outs\space\space
                   LAST-EXEC:\the\c@@@last@exec\space\space
                   TOPMARK:\topmarks\@@@out@mark\space\space
                   FIRSTMARK:\firstmarks\@@@out@mark\space\space
                   BOTMARK:\botmarks\@@@out@mark}%
```

Last valid pending entry = `\@tempcntb` = PENDING-OUTS − TOTAL-OUTS + BOTMARK
Number of invalid (moved) out's = `\@tempcnta` = TOTAL-OUTS − BOTMARK

```
    \advance\@tempcnta-\c@@@total@outs \advance\@tempcntb\@tempcnta
    \@tempcnta-\@tempcnta%
    \ifnum\@tempcnta>\z@
      \LL@debug@info{%
        >>> resave \the\@tempcnta\space output command(s).
            Too early to execute! <<<}%
    \fi
    \@tempcnta\z@ \def\reserved@b{}%
    \set@display@protect
    \@for\reserved@a :=\@@@pending@outs\do{%
      \ifx\reserved@a\@empty\else
        \ifnum\@tempcnta<\@tempcntb%
          \reserved@a%  execute output's related to the current page box.
          \global\advance\c@@@last@exec\@ne
          \LL@debug@info{>>> execute output command number
                   \the\c@@@last@exec\space<<<}%
        \else
```

Other output's must be resaved and still pending.

```
          \expandafter\g@addto@macro\expandafter\reserved@b\expandafter{%
            \reserved@a,}%
        \fi
        \advance\@tempcnta\@ne%
      \fi}%
    \expandafter\@temptokena\expandafter{\reserved@b}%
    \xdef\@@@pending@outs{\the\@temptokena}%
  \endgroup}%

  \fi% end of eTeX test.
```

`\protected@write` Because our redefined `\write` is expandable we must protect it inside of LATEX's `\protected@write`.

```
\long\def\protected@write#1#2#3{%
  \begingroup
     \let\thepage\relax
     #2%
     \let\protect\@unexpandable@protect
     \edef\reserved@a{\noexpand\write#1{#3}}%
     \reserved@a%
  \endgroup
  \if@nobreak\ifvmode\nobreak\fi\fi}%
```

`\@begindvi` Delay `\@begindvi` to be executed until the real page output.

```
  \let\@begindvi\@empty
\fi% end of \ifLL@combine
```

`\LL@toomanypages` Now elsewhere ...

```
\LoadClass{article}
\RequirePackage{graphicx}
```

`\LL@pagesize@specials` Now the graphics driver is known. We will check the graphics driver and set special code for page dimensions and other.

```
\newcommand*\LL@pagesize@specials[2]{}
\@ifundefined{Gin@driver}{}{%
  \ifx\Gin@driver\@empty\else%
    \filename@parse{\Gin@driver}\@tempswafalse%
    \def\reserved@a{dvips}%
    \ifx\filename@base\reserved@a\@tempswatrue\fi%
    \def\reserved@a{dvipdfm}%
    \ifx\filename@base\reserved@a\@tempswatrue\fi%
    \if@tempswa
      \ClassInfo{leaflet}{Generating code for dvips}%
      \def\LL@pagesize@specials#1#2{%
        \@tempdima=#1\@tempdimb=#2 %
        \AtBeginDvi{\special{papersize=\the\@tempdima,\the\@tempdimb}}}%
    \fi
    \def\reserved@a{pdftex}%
    \ifx\filename@base\reserved@a
      \ClassInfo{leaflet}{Generating code for pdfTeX}%
      \def\LL@pagesize@specials#1#2{%
        \@tempdima=#1\@tempdimb=#2 %
        \pdfpagewidth\@tempdima\pdfpageheight\@tempdimb}%
    \fi
    \def\reserved@a{luatex}%
    \ifx\filename@base\reserved@a
      \ClassInfo{leaflet}{Generating code for LuaTeX}%
      \ifx\pdfpagewidth\@undefined
        \def\LL@pagesize@specials#1#2{%
          \@tempdima=#1\@tempdimb=#2 %
          \pagewidth\@tempdima\pageheight\@tempdimb}%
      \else
        \def\LL@pagesize@specials#1#2{%
          \@tempdima=#1\@tempdimb=#2 %
          \pdfpagewidth\@tempdima\pdfpageheight\@tempdimb}%
      \fi
    \fi
    \def\reserved@a{xetex}%
    \ifx\filename@base\reserved@a
      \ClassInfo{leaflet}{Generating code for XeTeX}%
      \def\LL@pagesize@specials#1#2{%
        \@tempdima=#1\@tempdimb=#2%
        \pdfpagewidth\@tempdima\pdfpageheight\@tempdimb}%
    \fi
    \def\reserved@a{vtex}%
    \ifx\filename@base\reserved@a
      \ClassInfo{leaflet}{Generating code for VTeX}%
      \def\LL@pagesize@specials#1#2{%
        \@tempdima=#1\@tempdimb=#2%
        \mediawidth\@tempdima\mediaheight\@tempdimb}%
    \fi
  \fi
}
```

`\LL@CmdIgnored`

```
\newcommand*\LL@CmdIgnored[1]{%
```

```
  \ClassWarning{leaflet}{%
    `\string#1' ignored}}
```

2.4 Changing the class properties

```
\setlength{\parskip}{1ex plus 2pt}
\@listi%
\setlength{\labelwidth}{\leftmargin}
\addtolength{\labelwidth}{-\labelsep}
\pagestyle{empty}
\headheight\z@
\headsep\z@
\footskip\z@
\marginparwidth\z@
\marginparsep\z@
\sloppy
\setcounter{secnumdepth}{0}
%\renewcommand\marginpar[2][]{\LL@CmdIgnored{\marginpar}}
\renewcommand\twocolumn[1][]{\LL@CmdIgnored{\twocolumn}}
\renewcommand\onecolumn{\LL@CmdIgnored{\onecolumn}}
\renewcommand\topfraction{0.7}
\renewcommand\bottomfraction{0.7}
\setlength{\textfloatsep}{10pt plus 4pt minus 3pt}
\setlength{\parindent}{\z@}
\setlength{\leftmargini}{1.5em}
\setlength{\leftmarginii}{1.5em}
\setlength{\leftmarginiii}{1.5em}
\setlength{\leftmarginiv}{1.5em}
\setlength{\leftmarginv}{1.5em}
\setlength{\leftmarginvi}{1.5em}
\setlength{\labelsep}{.5em}
\setlength  \labelwidth{\leftmargini}
\addtolength\labelwidth{-\labelsep}
```

`\noparskip`

```
\def\noparskip{\par\vspace{-\parskip}}
```

Modification of list and section parameters for a more compact layout.

```
\let\old@small\small
\renewcommand{\small}{\old@small\let\@listi\@listI}
\let\old@footnotesize\footnotesize
\renewcommand{\footnotesize}{\old@footnotesize\let\@listi\@listI}
```

`\sectfont` User macro for changing the font of the typesetted parameter of the `\?section` commands.

```
\newcommand{\sectfont}{\bfseries}

\renewcommand\section{\@startsection{section}{1}{\z@}%
  {-3.5ex \@plus -.75ex}%
  {1ex} %{1.5ex}%
  {\normalfont\large\sectfont}}
\renewcommand\subsection{\@startsection{subsection}{2}{\z@}%
  {-2.5ex plus -.5ex}%
  {1\p@} %{1ex}%
  {\normalfont\normalsize\sectfont}}
\renewcommand\subsubsection{\@startsection{subsubsection}{3}{\z@}%
  {-2.5ex plus -.5ex}%
  {-1em}%
  {\normalfont\normalsize\sectfont}}
\def\part{\LL@CmdIgnored{\part}\secdef\@part\@spart}
\def\@part[#1]#2{}
\def\@spart#1{}

\renewcommand*\descriptionlabel[1]{%
  \hspace\labelsep\normalfont\descfont #1}
```

\descfont User macro for changing the font of the description label.

```
\newcommand*\descfont{\bfseries}

\iffalse
% Doesn't work!
\g@addto@macro\enumerate{\parsep2\p@\@plus2\p@\@minus\z@}
\g@addto@macro\itemize{\parsep2\p@\@plus2\p@\@minus\z@}
\g@addto@macro\description{\parsep2\p@\@plus2\p@\@minus\z@}
\else
\newcommand*\LL@listsetup{%
%  \parsep.445ex\@plus.445ex\@minus.2225ex%
%  \parsep.5ex\@plus.5ex\@minus.25ex%
  \parsep1ex\@plus.5ex\@minus.25ex%
  \LL@debug@info{***parsep=\the\parsep}%
%  \itemsep.25\parsep
  \itemsep\z@
  \LL@debug@info{***itemsep=\the\itemsep}%
  \topsep\z@
  \LL@debug@info{***topsep=\the\topsep}%
%  \partopsep\z@
  \LL@debug@info{***partopsep=\the\partopsep}%
}
\def\enumerate{%
  \ifnum \@enumdepth >\thr@@\@toodeep\else
    \advance\@enumdepth\@ne
    \edef\@enumctr{enum\romannumeral\the\@enumdepth}%
    \expandafter
    \list
      \csname label\@enumctr\endcsname
      {\usecounter\@enumctr
        \def\makelabel##1{\hss\llap{##1}}%
        %\def\makelabel##1{##1\hfill}%
        %\def\makelabel##1{\hss##1}%
        \LL@listsetup
      }%
  \fi}
\def\itemize{%
  \ifnum \@itemdepth >\thr@@\@toodeep\else
    \advance\@itemdepth\@ne
    \edef\@itemitem{labelitem\romannumeral\the\@itemdepth}%
    \expandafter
    \list
      \csname\@itemitem\endcsname
      {%
        \def\makelabel##1{\hss\llap{##1}}%
        %\def\makelabel##1{##1\hfill}%
        %\def\makelabel##1{\hss##1}%
        \LL@listsetup
      }%
  \fi}
\renewenvironment{description}
               {\list{}{\labelwidth\z@ \itemindent-\leftmargin
                        \let\makelabel\descriptionlabel
                        \LL@listsetup}}
               {\endlist}
\fi
```

2.5 Changing the page dimensions

\setmargins User command to specify the margins. (o, u, l, r)

\setmargins{⟨*top*⟩}{⟨*bottom*⟩}{⟨*left*⟩}{⟨*right*⟩}

The new user interface! By default, the leaflet class does not use headings, footings and marginal notes. If the user desires, these can be restored by setting the respective lengths *before*

calling \setmargins. Setting the margins should *always* be done using \setmargins; if the user assigns to any of the length registers \topmargin, \evensidemargin, \oddsidemargin using \setlength, chaos will happen.

```
\newcommand*\setmargins[4]{%
```

temp. top!

```
  \setlength\topmargin{#1}%
  \edef\LL@tmargin{\the\topmargin}%
```

temp. bottom!

```
  \setlength\evensidemargin{#2}%
  \textheight=\dimexpr\paperheight
    -\topmargin-\evensidemargin-\headheight-\headsep-\footskip
```

temp. left!

```
  \setlength\oddsidemargin{#3}%
```

temp. right!

```
  \setlength\evensidemargin{#4}%
  \textwidth=\dimexpr\paperwidth
    -\oddsidemargin-\evensidemargin-\marginparwidth-\marginparsep
```

real top!

```
  \addtolength\topmargin{-1in}%
```

real left!

```
  \addtolength\oddsidemargin{-1in}%
```

probably not necessary

```
  \evensidemargin\oddsidemargin
}
```

The dimensions of a single small page.

```
\LL@setPaperSize
\paperwidth=0.333333334\paperwidth
\setmargins{11mm}{11mm}{8mm}{8mm}
```

2.6 Additional user macros

\foldmarkrule
\foldmarklength

The dimension of the foldmark between page 2 and page 3.

```
\newcommand*\foldmarkrule{0.4pt}
\newcommand*\foldmarklength{2mm}
```

\AddToBackground
\AddToBackground*

Adds the given parameter #2 (some picture commands) to a macro which will be used inside a zero-dimensional picture environment located at the lower left corner of a page. Parameter #1 selects the page number. The star version adds the picture commands to the background of one of the combined large pages instead of one of the small pages.

```
\newcommand\AddToBackground{%
  \@ifstar{\@tempswatrue\LL@AddToBackground}
          {\@tempswafalse\LL@AddToBackground}}
\@onlypreamble\AddToBackground
\newcommand\LL@AddToBackground[2]{%
  \if@tempswa\def\@tempa{LL@largePic}\else\def\@tempa{LL@smallPic}\fi
  \expandafter\providecommand\csname\@tempa\@Roman{#1}\endcsname{}%
  \expandafter\g@addto@macro\csname\@tempa\@Roman{#1}\endcsname{#2}}
```

\CutLine
\CutLine*
\Scissors

Adds a line with two scissors on the left side of the page given as parameter #1. The starred version draws only the line.

```
\newcommand*\CutLine{%
  \@ifstar{\@tempswatrue\LL@CutLine}{\@tempswafalse\LL@CutLine}}
\@onlypreamble\CutLine
\newcommand*\LL@CutLine[2][.]{%
  \ifLL@combine
    \ifx\Scissors\@empty\@tempswatrue\fi
```

```
    \if .#1%
      \def\@tempa{\dotfill}%
    \else\if -#1%
      \def\@tempa{\hrulefill}%
    \fi\fi
    \edef\@tempb{%
      \noexpand\AddToBackground{#2}{%
        \noexpand\put(0,0){%
          \noexpand\rotatebox{90}{\noexpand\makebox(\paperheight,0){%
            \noexpand\normalsize\@tempa\if@tempswa\else\noexpand\Scissors%
              \@tempa\@tempa\noexpand\Scissors\@tempa\fi}}}}%
    }%
    \@tempb
  \fi}
\IfFileExists{pifont.sty}
  {\RequirePackage{pifont}%
   \newcommand*\Scissors{\raisebox{-0.85ex}{\large\ding{34}}}}%
  {\newcommand*\Scissors{}}
```

Adds the fold mark code between page 2 and page 3.

```
\AddToBackground{3}{\LL@foldmark}
```

2.7 Page construction and output

2.7.1 Intercepting the small pages

`\vb@xt@` Similar to `\hb@xt@` (should be part of the LaTeX kernel).

```
\providecommand*\vb@xt@{\vbox to}
```

We use our own page saving macro instead of real LaTeX output.

```
\AtBeginDocument{\AddToHook{shipout/before}{\LL@processPage}}
```

`LL@page` A counter for the small pages (absolute counting).

```
\newcounter{LL@page}\setcounter{LL@page}{1}
\newcommand\LL@tempa{}
```

`\LL@processPage` With "combine" option the six small pages will be saved in box registers (`\LL@boxI`...`\LL@boxVI`). The `\ShipoutBox` with the page content must be a bit moved to get a "normal-sized" box. In "nocombine" mode only put the background picture.

```
\newcommand*\LL@processPage{%
  \ifLL@combine\DiscardShipoutBox\fi
  \ifnum\c@LL@page<7\relax
    \LL@tempdima=\paperwidth
    \ifLL@combine
      \iffoldcorr
        \@tempswafalse
        \ifnum\c@LL@page=4\relax
          \@tempswatrue
        \else
          \ifnum\c@LL@page=5\relax
            \@tempswatrue
          \fi
        \fi
        \if@tempswa\LL@tempdima=0.9797979798\paperwidth
        \else\LL@tempdima=1.0101010101\paperwidth \fi
      \fi
      \@@@exec@outs
      \expandafter\newsavebox\csname LL@box\Roman{LL@page}\endcsname%
      \setbox\ShipoutBox=\vbox{\vskip1in\unvbox\ShipoutBox}%
      \setbox\ShipoutBox=\vbox{\moveright1in\box\ShipoutBox}%
      \setbox\ShipoutBox=\hb@xt@\paperwidth{\box\ShipoutBox\hss}%
      \iffoldcorr
        \setbox\ShipoutBox=\hb@xt@\LL@tempdima{\hss\box\ShipoutBox\hss}\fi
```

Add a background picture to a single small page ('combine').

```
      \setbox\ShipoutBox=\vb@xt@\paperheight{%
        \vbox{%
          \pictur@(0,0)(0,\paperheight)%
            \begingroup
              \set@typeset@protect
              \paperwidth=\LL@tempdima
              \@nameuse{LL@smallPic\Roman{LL@page}}%
            \endgroup
          \endpicture
        }%
        \nointerlineskip\box\ShipoutBox\vss
      }%
      \global\expandafter\setbox
        \csname LL@box\Roman{LL@page}\endcsname=\box\ShipoutBox
      \typeout{\@spaces[\the\c@LL@page] ==> [\Roman{LL@page}]}%
    \else
```

Add a background picture to a single small page ('nocombine').

```
      \setbox\ShipoutBox\vbox{%
        \vbox{\@tempdima=1in\relax
          \@tempdimb=\paperheight\advance\@tempdimb-\@tempdima
          \pictur@(0,0)(\@tempdima,\@tempdimb)%
            \begingroup
              \set@typeset@protect
              \@nameuse{LL@smallPic\Roman{LL@page}}%
            \endgroup
          \endpicture
        }%
        \nointerlineskip\box\ShipoutBox
      }%
    \fi
  \fi
  \ifnum\c@LL@page=7\relax
```

Make this an error or warning message, depending on class option. (`-->` warning in all cases?)

```
    \begingroup
      \set@typeset@protect
      \LL@toomanypages{%
        The text you supplied fills more than six pages\MessageBreak
        and will therefore not fit onto a single flyer}{%
        Try using smaller fonts or reducing vertical space}%
    \endgroup
  \fi
  \stepcounter{LL@page}}
```

NOTE TO OURSELVES: All the following must be done if the document code is already processed.

`\LL@add@empty@pages` Generating missing small pages

```
\newcommand*\LL@add@empty@pages{%
  \clearpage\pagestyle{empty}%
  \loop\ifnum\c@LL@page<7\relax
    \ClassInfo{leaflet}{Generating empty page \the\c@page}%
    \null\newpage
  \repeat
}
```

`LL@shipout@combined@pages`

```
\newcommand*\LL@shipout@combined@pages{%
```

2.7.2 Reassembling the small pages

Suppresses all shipout hook's clients (removing other kinds of background pictures). Our page saving macro is still needed for the next step and must be set again.

```
  \RemoveFromHook{shipout/foreground}[*]
  \RemoveFromHook{shipout/background}[*]
  \RemoveFromHook{shipout/lastpage}[*]
  \RemoveFromHook{shipout/before}[*]
  \AddToHook{shipout/before}{\LL@processPage}
```

Create empty pages if necessary.

```
  \LL@add@empty@pages
```

Here begins the real output. LaTeX's `\shipout` will no more suppressed. The begin dvi hook is reactivated. Final removing content from the hook.

```
  \RemoveFromHook{shipout/before}[*]
  \let\@begindvi\LL@begindvi
  \paperwidth=3\paperwidth
  \iflandscape
    \LL@pagesize@specials{\paperwidth}{\paperheight}%
  \else
    \LL@pagesize@specials{\paperheight}{\paperwidth}%
  \fi
```

`\LL@shipoutPage` This is similar to LaTeX's `\@outputpage`, but without a group; `\set@display@protect` seems to work as well.

`\LL@preparePages` `\LL@preparePage` Helper macros for page output: reassemble the small pages.

```
  \newcommand*\LL@shipoutPage[1]{%
%     \set@display@protect
    \let \protect \noexpand
    \shipout\vb@xt@\paperheight{%
      \set@typeset@protect
      \vskip-1in%
      \@begindvi\hb@xt@\paperwidth{\hskip-1in##1\hss}\vss}}%
  \newcommand*\LL@preparePages[3]{%
    \typeout{[\@Roman{##1}\space\@Roman{##2}\space\@Roman{##3}] ==>}%
```

Add a background picture to a large page.

```
    \pictur@(0,0)\@nameuse{LL@largePic\Roman{page}}\endpicture%
    \LL@preparePage{##1}\LL@preparePage{##2}\LL@preparePage{##3}}%
  \newcommand*\LL@preparePage[1]{%
    \expandafter\box\csname LL@box\@Roman{##1}\endcsname}%
```

Combine and output the frontside and the backside. The option two part arranges the pages in a different than the "natural" order, to allow the typesetting of a 4 page leaflet and a detachable 2 page form, for summaries, fill-in forms, applications, questionnaires, etc.

```
  \LL@selectOutput
  \iftwopart
    {\setcounter{page}{1}%
     \LL@shipoutPage{\LL@rotate@I{\LL@preparePages{6}{4}{1}}}}%
    {\setcounter{page}{2}%
     \LL@shipoutPage{\LL@rotate@II{\LL@preparePages{2}{3}{5}}}}%
  \else
    {\setcounter{page}{1}%
     \LL@shipoutPage{\LL@rotate@I{\LL@preparePages{5}{6}{1}}}}%
    {\setcounter{page}{2}%
     \LL@shipoutPage{\LL@rotate@II{\LL@preparePages{2}{3}{4}}}}%
  \fi
}%

\ifLL@combine
  \AddToHook{enddocument/end}{\LL@shipout@combined@pages}
\else
```

Create empty pages if necessary and then call the normal LaTeX end.

```
  \LL@pagesize@specials{\paperwidth}{\paperheight}%
  \AtEndDocument{%
```

```
    \LL@add@empty@pages
  }
\fi
⟨/class⟩
```

Done. Phew! (Did you hear that deep-drawn sigh?)

Change History

v0.8c
General: Repacked as dtx file. (RN) 1
v0.9d
General: LPPL 1.3 (HjG) 1
M-x delete-trailing-whitespace (HjG) . 1
More code documentation; some caveats. (HjG) 1
v0.9e
General: Add option 'finish'. (RN/HjG) . 1
v0.9f
General: Merge v0.9d and v0.9e (hastily). (HjG) 1
v0.9g
General: Rename option 'finish' to 'combine'. (RN/HjG) 1
v1.0
General: CTAN release. (RN/HjG) 1
v1.0a
General: Option 'debug'. (RN) 1
v1.0b
\closeout: \special must be processed similar to e. g. \write (VTEX). (RN) . 3
General: New option nospecialtricks for testing purposes. (RN) 1
v1.0c
\LL@shipoutPage: Fix \protect-ion problems. (RN/HjG) 12
v1.0d
\LL@foldmark: Invalid code for option "nofoldmark" changed. Thanks to Una Smith. (RN) 2
General: Options 'twopart/notwopart' suggested by Luis Rivera. No more 'final' in \ExecuteOptions suggested by Axel Berger (RN) 1
v1.0e
\LL@pagesize@specials: Patch from Toby Thurston for XeLaTeX. (RN) . . 6
v1.0pre1
General: Redefinition of \immediate, \write, \openout and \closeout. (RN) 1
v1.0pre2
General: eTeX support. (RN) 1
v1.0pre3
General: Options 'landscape' and 'portait'. (RN) 1
v1.1a
\LL@processPage: Additions for new option 'foldcorr'. (RN) 10
General: New options 'foldcorr/nofoldcorr' suggested by Walter Schmidt. (RN) 2
Using \dimexpr instead of package 'calc'.(RN) 6
v1.1b
\LL@pagesize@specials: Adding code for LuaTeX >0.85. Thanks to Knut Lickert for his bug report. (RN) 6
v2.0a
\LL@processPage: Do not call \@@@exec@outs temporarily. (RN) . . 10
Using LaTeX's new 'shipout/before' hook. (RN) 10
General: The macro \LenToUnit is no longer required. (RN) 1
v2.0b
\LL@processPage: Renamed \LL@savePage to \LL@processPage. (RN) 10
\LL@shipout@combined@pages: Suppress other background images. (RN) 11
General: Using hooks instead of redefinition of \@@end. (RN) 11
v2.0c
\@@@exec@outs: Missing \set@display@protect (RN) 5
\LL@processPage: Activated \@@@exec@outs again. (RN) 10
v2.0d
General: Using UTF8 in the preamble for author names to prevent unpacking issue. (RN) 1
v2.1a
\CutLine: Optional parameter for line type 9

Index

Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

Symbols

\@@@closeout 33, 102, 119, 146, 167

\@@@exec@outs . . . . . .

. . . . . 34, 147, 168, 450
\@@@immediate 32, 100, 115, 117, 119, 121, 140, 143, 146, 159, 163, 167
\@@@openout 33, 102, 117, 143, 163
\@@@out@mark . . 153, 158, 162, 166, 171, 176–178
\@@@pending@outs . . 100, 140, 143, 146, 148, 159, 163, 167, 188, 202
\@@@special 34, 101, 112, 121
\@@@write 32, 101, 115, 140, 159
\@Roman . . 402, 403, 527, 531
\@begindvi 11, 214, 512, 525
\@dummy@whatsit 125, 139, 142, 145
\@enumctr 346, 349, 350
\@enumdepth 344–346
\@for 188
\@ifequalsign 160
\@ifl@t@r 2
\@itemdepth 358–360
\@itemitem 360, 363
\@listI 301, 303
\@listi 272, 301, 303
\@minus . . . 326–328, 331–333
\@onlypreamble . . . 399, 406
\@part 317, 318
\@spart 317, 319
\@startsection 305, 309, 313
\@tempa 401–403, 411, 413, 419, 420
\@tempb 415, 422
\@toodeep 344, 358
\@unexpandable@protect 209

A

\AddToBackground 396, 416, 428
\AddToBackground* 396
\AddToHook . . . 430, 509, 546
\AtBeginDvi 230
\AtEndOfClass . . . 30, 63, 69

B

\botmarks 171, 178
\bottomfraction 287

C

\c@@@last@exec 170, 175, 192, 194
\c@@@total@outs 157, 158, 161, 162, 165, 166, 170, 174, 179
\c@LL@page 435, 440, 443, 471, 487, 499
\c@page 500
\ClassWarning 269
\closeout 33, 100
\CutLine 404
\CutLine* 404

D

\descfont 322, 323
\description 328
\descriptionlabel 321, 373
\dimexpr 53, 381, 385
\ding 426
\DiscardShipoutBox 434
\do 188
\dotfill 411

E

\endlist 375
\endpicture . . . 465, 481, 528
\enumerate 326, 343
\evensidemargin 380, 382, 384, 386, 389

F

\filename@base 223, 225, 233, 240, 253, 260
\filename@parse 221
\firstmarks 177
\fmtversion 2
\foldcorrfalse 60
\foldcorrtrue 59
\foldmarklength . . . 55, 394
\foldmarkrule 52, 394
\footnotesize 302, 303
\footskip 278, 382

G

\Gin@driver 220, 221
\globmarks 153

H

\headheight 276, 382
\headsep 277, 382
\hrulefill 413
\hskip 525
\hss 351, 353, 365, 367, 454, 456, 525

I

\if@nobreak 213
\iffalse 324
\iffoldcorr 15, 438, 455
\IfFormatAtLeastTF 2, 3
\iflandscape . . . 12, 85, 514
\ifLL@combine 31, 39, 70, 83, 215, 408, 434, 437, 545
\iftumble 13, 87, 94
\iftwopart 14, 533
\ifvmode 213
\immediate 32, 100
\immediate@closeout 106, 118
\immediate@openout 105, 116
\immediate@special 107, 120
\immediate@write . 104, 114
\itemindent 372
\itemize 327, 357
\itemsep 335–337

L

\labelsep 274, 296, 298, 322
\labelwidth 273, 274, 297, 298, 372
\landscapefalse 25
\landscapetrue 24
\leftmargin 273, 372
\leftmargini 290, 297
\leftmarginii 291
\leftmarginiii 292
\leftmarginiv 293
\leftmarginv 294
\leftmarginvi 295
\line 55
\linethickness 52
\list 348, 362, 372
\LL@add@empty@pages 497, 510, 550
\LL@AddToBackground 397, 398, 400
\LL@begindvi 11, 512
\LL@CmdIgnored 268, 283–285, 317
\LL@combinefalse 41
\LL@combinetrue 40
\LL@CutLine 405, 407
\LL@debug@info 17, 26, 149, 173, 182, 193, 334, 337, 339, 341
\LL@foldmark 48, 428
\LL@listsetup 330, 354, 368, 374
\LL@page 431
\LL@pagesize@specials 218, 515, 517, 548
\LL@preparePage 519
\LL@preparePages 519, 535, 537, 540, 542
\LL@processPage 430, 433, 509
\LL@rotate@I . . . 84, 535, 540
\LL@rotate@II . . 84, 537, 542
\LL@selectOutput . . 42, 532
\LL@setPaperSize . . 36, 391
\LL@shipout@combined@pages 504, 546
\LL@shipoutPage 519, 535, 537, 540, 542
\LL@tempa 432
\LL@tempdima 16, 436, 447, 448, 456, 462
\LL@tmargin 53, 379
\LL@toomanypages 61, 64, 71, 74, 216, 490
\llap 351, 365
\long 113, 138, 156, 205

M

\makelabel 351–353, 365–367, 373
\marginpar 283
\marginparsep 280, 386
\marginparwidth . . 279, 386
\marks 158, 162, 166
\mediaheight 264
\mediawidth 264
\moveright 453

N

\newcounter . . . 154, 155, 431
\newlength 16
\nobreak 213
\nointerlineskip . 467, 483
\noparskip 299
\null 501
\number 140, 143, 146, 159, 163, 167

O

\oddsidemargin 383, 386, 388, 389
\old@footnotesize 302, 303
\old@small 300, 301
\onecolumn 285
\openout 33, 100

P

\pageheight 245
\pagestyle 275, 498
\pagewidth 245
\paperheight 38, 53, 381, 418, 457, 459, 475, 515, 517, 522, 548
\paperwidth 38, 385, 392, 436, 447, 448, 454, 462, 513, 515, 517, 525, 548
\part 317
\partopsep 340, 341
\PassOptionsToClass 62, 68, 79, 80
\PassOptionsToPackage 18–21
\pdfliteral 129
\pdfoutput 129
\pdfpageheight 237, 249, 257
\pdfpagewidth 237, 242, 249, 257
\pending@@@openout 160
\pending@@openout 160
\pending@@write 113, 138, 156
\pending@closeout 111, 124, 144, 164
\pending@openout 110, 123, 141, 160
\pending@write 109, 113, 122
\pictur@ 459, 476, 528
\protected@write 205

R

\raisebox 426
\RemoveFromHook 505–508, 511
\reserved@b . . . 186, 196, 201
\reset@immediate 108, 115, 117, 119, 121
\rotatebox 88, 93, 95, 97, 418

S

\Scissors 404
\secdef 317
\sectfont 304, 308, 312, 316
\set@display@protect 187, 520
\set@typeset@protect 461, 478, 489, 523
\setmargins 377, 393
\shipout 522
\ShipoutBox 452–454, 456, 457, 467, 470, 473, 483
\sloppy 281
\special 34, 101, 107, 112, 125, 230
\stepcounter 496
\subsubsection 313

T

\textfloatsep 288
\textheight 381
\textwidth 385
\thepage 207
\topfraction 286
\topmargin 378, 379, 382, 387
\topmarks 176
\topsep 338, 339
\tumblefalse 47
\tumbletrue 46
\twocolumn 284
\twopartfalse 28
\twoparttrue 27

U

\unvbox 452

V

\vb@xt@ 429, 457, 522
\vskip 452, 524
\vss 467, 525

W

\write 32, 100, 210